

VENKAT KOUSHIK PILLALA

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Portfolio | LinkedIn | GitHub

SUMMARY

AI/ML Engineer with an M.S. in Applied Artificial Intelligence from Stevens Institute of Technology, focused on LLM evaluation, RAG systems, and multimodal AI. Brings an evaluation-first approach shaped by building retrieval and multimodal evaluation workflows, along with professional experience in domain-specific LLM experimentation and NLP model development. Hands-on with fine-tuning, retrieval pipelines, and AWS-based ML deployment.

TECHNICAL SKILLS

Programming & APIs: Python, SQL, C++, FastAPI, REST APIs, Linux/Bash, Git

Machine Learning & NLP: PyTorch, Hugging Face Transformers, Scikit-learn, TensorFlow, XGBoost, NLP, Text Classification, Computer Vision, Model Evaluation

Generative AI & Retrieval: LLMs, Retrieval-Augmented Generation (RAG), LangChain, LangGraph, LoRA Fine-Tuning, Prompt Engineering, LLM Evaluation, Embeddings, Reranking

Vector Databases & Data Systems: FAISS, ChromaDB, LanceDB, PostgreSQL, MongoDB, BigQuery, Apache Spark

Cloud & MLOps: AWS SageMaker, Amazon S3, AWS Glue, AWS Lambda, Amazon EC2, Amazon Rekognition, Amazon Comprehend, Docker, MLflow, CI/CD, Model Deployment

PROFESSIONAL EXPERIENCE

Aspire Technolab – ML Engineer Intern

Jan. 2024 – May 2024

Technologies: Python, PyTorch, Hugging Face Transformers, LLaMA 2, LoRA, NLP, OpenCV, CNNs, REST APIs

Bengaluru, India

- Supported domain-specific LLM experimentation using LLaMA 2 and LoRA, performing tokenization, prompt testing, benchmark execution, and model-output review for enterprise-oriented use cases.
- Developed NLP classification pipelines for sentiment analysis and spam detection using PyTorch and Hugging Face Transformers, evaluating performance through precision, recall, F1-score, and confusion matrices.
- Achieved a **0.92 F1-score** on internal NLP evaluation tasks by improving preprocessing, comparing model configurations, and conducting recurring error analysis.
- Built CNN-based visual inspection experiments using OpenCV preprocessing and model evaluation workflows to identify image-classification failure patterns.

FunMotionLabs – Game Development Intern, AI/ML Systems

Jan. 2023 – Dec. 2023

Technologies: Python, NLP, Machine Learning, Reinforcement Learning, Unity, C#, FSM

Hyderabad, India

- Developed intelligent agent systems using machine learning, heuristic decision logic, and Finite State Machines to enable adaptive behavior and real-time environment-aware decisions.
- Built NLP-based interaction workflows to interpret user inputs, identify intent-driven triggers, and map them to context-aware agent responses and actions.
- Applied reinforcement learning concepts and simulation-based testing to improve autonomous navigation, behavioral consistency, and response selection across interactive scenarios.
- Delivered reusable AI interaction and behavior components for the Kurenai module, supporting integration into the broader application workflow.

Credo Systemz – AWS ML Developer Trainee

July 2022 – Oct. 2022

Technologies: AWS SageMaker, Amazon S3, AWS Glue, AWS Lambda, EC2, Docker

Chennai, India

- Built cloud-based machine learning workflows on AWS SageMaker, covering Amazon S3 data ingestion, feature preparation, model training, evaluation, and endpoint deployment.
- Developed ETL and feature engineering pipelines using AWS Glue and SageMaker Data Wrangler to prepare structured training inputs for downstream ML workflows.
- Deployed model inference workflows using SageMaker endpoints, AWS Lambda, Amazon EC2, Docker, and REST APIs to support scalable model testing and application integration.
- Implemented applied AI workflows using Amazon Rekognition for image analysis and Amazon Comprehend for NLP-based entity extraction from unstructured data.

PROJECTS

Secure Local RAG and LLM Evaluation Platform

Feb. 2026 – April 2026

Python, LLaMA 3, LangChain, ChromaDB, FastAPI, Docker, SQL

- Built a local RAG platform for semantic search and question answering over sensitive documents and codebases without external API reliance.
- Implemented ingestion, vector retrieval, SQL-backed evaluation, and failure analysis workflows, reducing hallucinated responses by **34%**.
- Deployed a Dockerized FastAPI inference service with SQL-backed metadata tracking for repeatable local evaluation and auditability.

Biomedical Multimodal RAG Evaluation System

Nov. 2025 – Jan. 2026

Python, IDEFICS2-8B, OpenCLIP, MiniLM, ChromaDB, PyTorch

- Designed a multimodal RAG pipeline combining biomedical text and medical images through MiniLM and OpenCLIP embeddings in ChromaDB.
- Integrated a 4-bit quantized IDEFICS2-8B model and evaluated retrieval quality, answer faithfulness, and hallucination failure patterns.
- Built a failure-analysis workflow to compare text-only and image-grounded retrieval results across multimodal query scenarios.
- Achieved **84% Recall@5** across **75 biomedical queries**, improving multimodal retrieval by **13%** over text-only search.

EDUCATION

Stevens Institute of Technology

Hoboken, NJ

Master of Science in Applied Artificial Intelligence | **CGPA: 3.76**

August 2024 – May 2026

Relevant Coursework: Machine Learning; Deep Learning; Big Data Analytics; Pattern Recognition & Classification; Probability & Stochastics; Data Structures & Algorithms

Sathyabama University

Chennai, India

B.E. in Computer Science Engineering | **CGPA: 8.7**

August 2020 – May 2024

LEADERSHIP & INVOLVEMENT

- Led AWS Cloud Club and AI-UI/UX Club activities focused on cloud, AI, and product design.
- Coordinated Debate Club sessions, strengthening public speaking and structured communication.